17 · The three regions of the ear

The words of this track, as it says them. Bold brown is the examiner.

The three regions of the ear. Four and a half marks.

What are the three regions of the ear? Describe their structures and their functions.

Four and a half marks means about nine half-mark facts. Three regions, three facts each.

The hook: the outer ear collects, the middle ear amplifies, and the inner ear senses. And the inner ear does two jobs: the cochlea hears, and the vestibular apparatus balances.

The outer ear, or external ear. The pinna, the external auditory canal, and the tympanic membrane, which is the eardrum. Glands in the canal make cerumen, which is earwax. Its function: it collects sound and directs it onto the eardrum, and the pinna also helps you locate where a sound is coming from.

The middle ear. An air-filled cavity in the temporal bone, holding the ossicles: the malleus, the incus and the stapes. Its function: it amplifies sound, passing it from the large eardrum to the much smaller oval window. It also has the Eustachian tube, also called the pharyngotympanic tube, running to the nasopharynx, which equalises pressure. And two small muscles, the tensor tympani and the stapedius, which dampen loud sounds.

The inner ear. The bony labyrinth in the temporal bone, filled with perilymph, and inside it the membranous labyrinth, filled with endolymph. It has two parts. The cochlea, for hearing: its organ of Corti turns sound vibration into nerve impulses. And the vestibular apparatus, which is the vestibule and the semicircular canals, for equilibrium, or balance.

The traps. Only the inner ear does balance. The outer and middle ear are for hearing only. Second: the Eustachian tube equalises pressure. It's why your ears pop on a plane. It doesn't carry sound. Third: perilymph is in the bony labyrinth, on the outside. Endolymph is inside the membranous labyrinth. Endo means inside.

Once more. The three regions of the ear.

The outer ear: the pinna, the external auditory canal with its cerumen, and the tympanic membrane. It collects sound onto the eardrum and helps locate it. The middle ear: air-filled, in the temporal bone, with the ossicles, which amplify sound onto the oval window. The Eustachian tube equalises pressure, and two small muscles dampen loud sounds. The inner ear: a bony labyrinth of perilymph around a membranous labyrinth of endolymph. The cochlea, with its organ of Corti, hears. The vestibular apparatus balances.